

Agentic WhatsApp Customer Engagement & Lead Automation Platform

Full-Stack Agentic AI · Generative AI · RAG · Voice & Media Intelligence · ERP / CRM Integration · Composable CMS & Marketing Cloud · WhatsApp Automation Platform

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Revision	v4 — Adds a Strapi-powered dynamic knowledge base, Adobe Experience Cloud (Marketo Engage / Adobe Campaign) syncing, proactive AI-initiated outbound campaigns, live call sentiment analysis with auto-escalation, and approved ERP write actions (leave, tickets, expense claims). v3 added Frappe CRM/ERPNext with OTP self-service; v2 added voice, media, HubSpot, and calendar booking.

This revision makes the platform proactive and two-way: a Strapi-managed knowledge base keeps answers current without re-uploading PDFs, qualified leads and behavior can flow into Adobe Marketo Engage or Adobe Campaign, the AI can reach out first (reminders, follow-ups), live calls are monitored for frustration and escalated automatically, and verified employees can now write to the ERP — not just read from it — for routine requests like leave and expense claims.

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Tags: **NEW** = added in this revision (v4) · **UPDATED** = existing content changed in v4.

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1. Project Overview

The Agentic WhatsApp Customer Engagement & Lead Automation Platform is an AI-powered automation system that connects a business's WhatsApp communication channel — and, with this revision, its phone line as well — to an intelligent agent capable of understanding customers, answering questions, qualifying leads, and escalating to a human when necessary.

Rather than replying with fixed, scripted messages, the platform uses a Large Language Model (LLM) coordinated by an agentic workflow. The system can reason about what a customer wants, decide which specialized capability should handle the conversation, retrieve accurate business information, understand images, documents, and voice notes, take structured actions such as saving a lead or booking a calendar meeting, and hand off to a human agent — over chat or over a live phone call — when confidence is low or the customer explicitly asks for one.

Core capabilities

- Understand customer messages and identify intent (support question, sales inquiry, lead, complaint, etc.)
- Answer questions using the business's own knowledge base (RAG)
- Retrieve relevant, grounded information instead of relying purely on the model's general knowledge
- Understand voice messages, images, and documents sent by the customer
- Answer and hold live phone calls, listening and responding in real time
- Qualify potential customers and capture structured leads
- Execute controlled tools/actions (e.g., create a lead, book a calendar meeting, sync to HubSpot, send a webhook event)
- Maintain conversation context across multiple messages, calls, and sessions
- Escalate conversations — chat or voice — to a human agent and support full human takeover
- Serve verified employee and customer records (employee details, invoice status) live from the company's own self-hosted ERPNext, in chat, after OTP verification on WhatsApp
- Sync leads to the CRM of the business's choice — self-hosted Frappe CRM or HubSpot
- Keep its product, pricing, and FAQ knowledge current from a Strapi-managed content source, without re-uploading documents **NEW**
- Push qualified leads and behavioral signals into Adobe Marketo Engage or Adobe Campaign for personalized follow-up **NEW**
- Reach out first — proactive WhatsApp messages and outbound calls for reminders, follow-ups, and ERP-triggered events **NEW**
- Detect caller frustration in real time during a phone call and escalate to a human automatically **NEW**
- Let a verified employee not just view but act — raising a leave application, expense claim, or support ticket in ERPNext from WhatsApp **NEW**
- Provide a dashboard for monitoring conversations, calls, leads, and AI decisions
- Continuously improve the knowledge base using unanswered-question tracking

Why this is more than a traditional chatbot

A traditional chatbot follows fixed decision trees or keyword-matched replies: it cannot reason about ambiguous input, cannot retrieve knowledge outside its scripted flow, and cannot decide on its own that

a task requires a different specialist or a human. This platform is built on an **Agentic AI architecture**, where a Supervisor Agent coordinates specialized agents, tool execution, knowledge retrieval, multimodal understanding, voice interaction, workflow state, and human intervention. The system does not just generate text — it listens, sees, plans, retrieves, decides, acts, and remembers, within a controlled and auditable framework.

2. Problem Statement

Businesses that rely on WhatsApp and phone calls for customer communication consistently run into the same operational problems as message and call volume grows:

- A large proportion of incoming messages and calls are repetitive, common questions
- Response times slow down as the team gets overwhelmed
- Promising leads get lost in the noise of routine conversations
- Employees spend significant time manually answering the same questions — by chat and by phone
- Different team members give inconsistent answers to the same question
- Business knowledge is scattered across PDFs, websites, brochures, and individual employees' memory
- Customers frequently send voice notes, photos, and scanned documents that go unread or take time to review manually
- Inbound calls are missed entirely outside business hours, losing leads
- Leads are not qualified consistently — critical details (budget, timeline, requirement) are missed, and rarely make it into the CRM automatically
- Meetings and demos require manual back-and-forth to schedule
- Employees repeatedly ask HR or admin for routine details (profile, leave balance, attendance) that already sit in the ERP
- Customers repeatedly ask for invoice status, amount due, and payment confirmation that already sit in the accounting system
- Personal and financial data cannot safely be shared over chat without a reliable identity check
- There is no centralized place to monitor what is happening across conversations and calls
- AI systems used in isolation sometimes hallucinate information not grounded in real business facts
- There is no structured path for a human to step in when the AI is uncertain
- Business owners have little to no visibility into what an AI system is doing or why

How this project addresses these problems

The platform addresses each of these issues directly: repetitive questions are automated through the Knowledge Agent and RAG pipeline; voice notes, images, and documents are automatically transcribed and understood instead of sitting unread; inbound phone calls are answered by the AI Voice Call Agent around the clock; leads are captured, qualified, and pushed straight into the CRM (Frappe CRM or HubSpot); meetings are booked automatically onto the business's calendar; a Human Escalation Agent and takeover system ensure a person can intervene at any point, on chat or on a call; and the React dashboard gives full visibility into conversations, calls, AI decisions, tool calls, and audit logs. Routine employee and customer look-ups are served directly from the company's own ERPNext after OTP verification on WhatsApp, so the data never leaves the company's infrastructure.

3. Project Objectives

The project is guided by the following objectives:

1. Automate repetitive customer communication across chat and voice.
2. Provide business-specific AI responses rather than generic replies.
3. Ground AI answers in trusted business knowledge (RAG).
4. Understand voice messages, images, and documents shared by customers.
5. Answer and manage live inbound phone calls with an AI voice agent.
6. Automate lead qualification.
7. Capture structured customer information and sync it automatically to the business's chosen CRM (Frappe CRM or HubSpot) for follow-up. **UPDATED**
8. Automate meeting and demo scheduling via calendar booking.
9. Give verified employees self-service access to their own ERPNext records over WhatsApp.
10. Give verified customers instant invoice and account information from ERPNext.
11. Verify identity with a WhatsApp OTP before any personal or financial data is shared.
12. Integrate with self-hosted Frappe CRM and ERPNext, keeping business data on the company's own server.
13. Provide human-in-the-loop escalation for complex or sensitive cases, on chat and on calls.
14. Provide a centralized dashboard for monitoring and management.
15. Enable businesses to customize their AI agent's persona and behavior.
16. Support multiple languages, including regional and mixed-language conversations.
17. Provide tool-based automation instead of free-text-only responses.
18. Maintain conversation context across turns, calls, and sessions.
19. Provide monitoring, logging, and auditing of system behavior.
20. Reduce hallucination through RAG and response validation.
21. Keep the knowledge base current and structured using a headless CMS rather than static document re-uploads.
22. Extend the business's CRM data into its marketing stack (Adobe Experience Cloud) for retargeting.
23. Move from reactive to proactive engagement with AI-initiated outreach.
24. Detect and act on customer frustration in live calls before it escalates.
25. Allow safe, approved write-back to the ERP for routine employee self-service requests.
26. Create an extensible, maintainable, multimodal agentic AI architecture.

4. Major Features

A. WhatsApp Integration

The architecture is designed around the official WhatsApp Business platform, ensuring compliance and long-term reliability rather than depending on unofficial or reverse-engineered clients.

- WhatsApp Business Cloud API for sending and receiving messages
- Webhook integration to receive real-time events from Meta
- Incoming message processing and normalization
- Outgoing AI-generated responses

-
- Message status handling (sent, delivered, read, failed)
 - Contact management (customer profile, phone number, metadata)
 - Conversation management (threading messages into conversations)
 - Customer opt-in / opt-out handling for compliant messaging
 - General WhatsApp event processing (template status, account alerts, etc.)

B. Voice, Image & Document Understanding

The platform is no longer limited to plain text. Every media type a customer sends through WhatsApp is captured, understood, and folded into the same agentic pipeline used for text.

- **Voice messages** — incoming WhatsApp voice notes are downloaded and transcribed to text, then processed exactly like a typed message so the full agent, RAG, and lead pipeline applies unchanged
- **WhatsApp media understanding** — images and documents attached to a WhatsApp conversation (photos, screenshots, PDFs, scanned forms) are automatically fetched and analyzed
- **Image / document understanding** — a vision-capable model extracts the meaning of the media (e.g., a product photo, a broken-item picture, a filled requirement form, an ID or invoice) and turns it into structured, usable context for the Supervisor Agent
- Spoken replies — outgoing AI responses can optionally be generated as a voice note using text-to-speech, matching the business's configured voice persona
- All media is stored and linked to its originating conversation and message record for full auditability

C. AI Voice Call Agent

Beyond WhatsApp, the platform can now answer and hold a real, live phone call.

- Inbound calls to the business number are answered automatically by the AI
- The AI **listens** to the caller in real time using streaming speech-to-text
- The same Supervisor Agent, knowledge base, and lead-qualification logic used for WhatsApp reasons over the live transcript
- The AI **responds** out loud using low-latency text-to-speech, holding a natural back-and-forth conversation
- Calls can be qualified into leads, answered from the knowledge base, or transferred live to a human when needed
- Every call is recorded, transcribed, and logged alongside the customer's WhatsApp history

D. CRM & Calendar Automation

- **Pluggable CRM integration — Frappe CRM or HubSpot** UPDATED — qualified leads captured from WhatsApp or from a phone call are automatically created or updated in the business's chosen CRM (Leads and Deals in self-hosted Frappe CRM, or Contacts and Deals in HubSpot), with the full conversation/call summary attached. Each workspace selects its provider (or both) in the dashboard.
- **Calendar booking automation** — the AI can check a connected calendar's real-time availability and book a meeting, demo, or call-back directly inside the conversation or phone call, without human involvement
- Booking confirmations are sent back to the customer over WhatsApp automatically

E. Enterprise Self-Service — Frappe CRM & ERPNext

The platform connects to the company's **own self-hosted** ERPNext and Frappe CRM servers and serves records live in the WhatsApp chat — but only after identity verification.

- **Employee self-service** — the employee types their Employee ID in WhatsApp; the system checks the ID against ERPNext, confirms the WhatsApp number matches the mobile number registered for that employee, sends an OTP on WhatsApp, and after successful verification shows the employee's details fetched from the server
- **Customer invoice lookup** — the customer enters an Invoice ID and their WhatsApp number; the system matches both against the invoice and customer record in ERPNext, sends an OTP on WhatsApp, and after verification shows the invoice details
- **Lead capture into Frappe CRM** — AI-qualified leads are created in the self-hosted Frappe CRM (or HubSpot, by choice)
- All lookups are read-only, limited to an allowlist of fields, audited, and restricted to the verified person's own records (see Sections 9 and 10)

F. Dynamic Knowledge Base via Strapi NEW

Instead of relying only on static PDF re-uploads, the knowledge base can be backed by **Strapi**, an open-source headless CMS. Business teams edit products, pricing, and FAQs in a familiar content-editor UI; the AI reads the current published content through Strapi's API in real time.

- Marketing or product teams update pricing/FAQ content in Strapi without involving engineering or re-uploading files
- The RAG pipeline treats Strapi as a live knowledge source alongside PDFs and URLs, re-indexing automatically on publish
- Draft content in Strapi is never surfaced to customers — only published entries are ingested
- Reduces knowledge staleness: a price change in Strapi is reflected in the AI's answers within minutes, not after a manual re-upload

G. Marketing Cloud Sync — Adobe Experience Cloud NEW

- Qualified leads and conversation-derived behavioral data (intent, engagement, topics discussed) are pushed to **Adobe Marketo Engage** or **Adobe Campaign**
- Enables automated, personalized email and ad-retargeting journeys triggered directly from a WhatsApp conversation or call
- Complements, rather than replaces, the Frappe CRM / HubSpot sync in Section 9.4 — the CRM remains the system of record for the sales pipeline, while Adobe Experience Cloud receives a copy for marketing automation

H. Proactive Outbound Campaigns NEW

- The AI can initiate contact instead of only responding — outbound WhatsApp messages and outbound voice calls
- Triggers include appointment reminders, invoice/payment follow-ups, and ERPNext events (e.g., leave approved, invoice overdue, delivery dispatched)
- All outbound campaigns respect WhatsApp opt-in/opt-out status and messaging-window rules (see Section 11.3)

I. Live Sentiment Analysis on Calls NEW

- Real-time sentiment tracking runs alongside the Voice Call Agent's speech-to-text stream
- A sustained negative trend (frustration, anger) automatically triggers escalation to a human — the same Human Escalation Agent and live-transfer mechanism used for low-confidence answers

J. ERP Write Actions NEW

- Moves selected ERP interactions from read-only to write, for a verified employee only
- Supported actions: raise a Leave Application, submit an Expense Claim, open a Support/Helpdesk ticket
- Every write action is a two-step confirm-then-submit flow with a plain-language summary shown before anything is written to ERPNext (see Section 11.5)

K. AI Agent System

The platform uses multiple specialized AI agents coordinated through a central Supervisor Agent, rather than a single monolithic prompt trying to do everything.

Supervisor Agent

Responsible for: understanding the overall request, intent detection, agent routing, workflow coordination across multiple steps, maintaining conversation state, tool selection, error handling, and escalation decisions — across both chat and voice channels.

Knowledge Agent

Responsible for: business information queries, RAG retrieval from the knowledge base, product / service information, FAQ handling, and producing grounded, source-backed answers.

Media Understanding Agent

Responsible for: transcribing voice notes, analyzing images and documents sent by the customer, and converting that media into structured context the rest of the system can act on.

Sales Agent

Responsible for: detecting purchase intent, understanding customer requirements, carrying sales-oriented conversations forward, and recommending products/services based on available business information.

Lead Qualification Agent

Responsible for extracting structured lead data, including: name, phone, email, company, requirement, service, budget, timeline, lead intent, and lead score — then syncing qualified leads to the configured CRM (Frappe CRM or HubSpot).

Support Agent

Responsible for: general customer support, FAQs, business information queries, and common customer issues.

Voice Call Agent

Responsible for: answering inbound phone calls, running real-time speech-to-text and text-to-speech, holding the live conversation, and handing the resulting transcript to the Supervisor Agent for the same intent detection, RAG, and lead-extraction pipeline used on WhatsApp.

Outbound Campaign Agent NEW

Responsible for: selecting the customers or employees due for a reminder, follow-up, or ERP-triggered notification, respecting opt-in and quiet-hours rules, and initiating the outbound WhatsApp message or voice call.

Sentiment Monitoring Agent NEW

Responsible for: scoring caller sentiment turn-by-turn during a live call from the streaming transcript and voice signal, and notifying the Human Escalation Agent when frustration crosses a configured threshold.

Scheduling Agent

Responsible for: checking calendar availability and booking meetings, demos, or call-backs on behalf of the business, and confirming the booking back to the customer.

Enterprise Data Agent

Responsible for: recognising employee-ID and invoice-ID requests, starting the verification flow, fetching only authorised fields from ERPNext through read-only connector tools once a verified session exists, and formatting the result for WhatsApp. It does **not** decide who is verified — that is enforced by the deterministic OTP Verification Service (Section 10), which is ordinary application code rather than an LLM decision.

Human Escalation Agent

Responsible for: detecting when the AI should stop responding automatically, handling low-confidence situations, handling explicit requests for a human, escalating complex or sensitive cases, and triggering human takeover of a conversation or a live call transfer.

5. Agentic AI Capabilities

This section explains, in concrete terms, why the platform qualifies as an Agentic AI system rather than a conventional chatbot or a single prompt-response wrapper around an LLM.

Intent Detection

The system determines what the customer is trying to accomplish before deciding how to respond — whether the input arrives as text, a transcribed voice note, an analyzed image, or a live call transcript.

Example — Customer (voice note): "I need an ecommerce website."

System: Transcribes voice note → Intent = Service Inquiry

Agent Routing

The Supervisor Agent decides which specialized agent should handle a given request.

```
Customer Query (text / voice note / image / live call)
↓
Supervisor
↓
Knowledge Agent
↓
RAG
```

↓
Response

Or, for a sales-oriented message:

Customer Query
↓
Supervisor
↓
Lead Agent
↓
Lead Extraction
↓
HubSpot CRM / Webhook

Or, for an inbound phone call:

Incoming Call
↓
Voice Call Agent (listens)
↓
Supervisor
↓
Knowledge / Lead / Scheduling Agent
↓
Voice Call Agent (responds)

Or, for an employee or customer record lookup:

Employee ID / Invoice ID in chat
↓
Supervisor
↓
Enterprise Data Agent
↓
OTP Verification Service (must succeed first)
↓
ERPNext connector (read-only, allowlisted fields)
↓
Verified details shown in WhatsApp

Tool Calling

Agents can use controlled tools instead of only generating free-form text. This allows the system to take real actions — storing data, sending messages, analyzing media, booking a meeting, or triggering escalation — in a predictable, auditable way.

Example tools:

- `search_knowledge()`
- `get_product_information()`
- `transcribe_voice_message()`

-
- analyze_image()
 - analyze_document()
 - create_lead()
 - update_lead()
 - sync_lead_to_crm() **UPDATED** — provider-routed to Frappe CRM and/or HubSpot
 - start_verification()
 - send_whatsapp_otp()
 - verify_otp()
 - get_employee_details() — verified session only
 - get_invoice_details() — verified session only
 - query_strapi_content() **NEW**
 - sync_to_adobe_experience_cloud() **NEW**
 - trigger_outbound_campaign() **NEW**
 - score_call_sentiment() **NEW**
 - submit_leave_application() **NEW** — verified session, confirm-then-submit
 - submit_expense_claim() **NEW** — verified session, confirm-then-submit
 - create_support_ticket() **NEW** — verified session, confirm-then-submit
 - check_calendar_availability()
 - book_calendar_event()
 - send_message()
 - answer_phone_call()
 - transfer_call_to_human()
 - escalate_to_human()
 - create_webhook_event()
 - get_customer_history()

Tool execution is validated and permission-controlled — an agent cannot invoke a tool it is not authorized to use, and every call passes through input validation before execution (see Section 17, Security and Governance).

Identity-bound tools: get_employee_details() and get_invoice_details() accept no employee or invoice identifier from the model. The subject is read from the server-side verified session, so a manipulated prompt cannot make the AI fetch someone else's record.

Stateful Workflows

LangGraph maintains explicit workflow state across the lifetime of a conversation or call, rather than treating each message as an isolated request. Tracked state includes:

- Conversation ID / Call ID
- Customer ID
- Current intent
- Current agent
- Retrieved knowledge
- Media analyzed so far (voice, image, document)
- Lead information
- Calendar booking status

-
- Verification status and verified subject (employee / customer)
 - Tool calls made so far
 - Escalation status
 - Final response

Human-in-the-Loop

```
AI detects need for human (chat or live call)
↓
Automation pauses / call is transferred
↓
Conversation or call assigned to human
↓
Human responds
↓
AI remains paused
↓
AI can later resume
```

6. Knowledge Base and RAG

Retrieval-Augmented Generation (RAG) is the mechanism the platform uses to ground AI responses in real, trusted business information rather than relying solely on the language model's general training data.

Supported knowledge sources

- PDF documents
- Website URLs
- Manually entered text
- FAQs
- Product information
- General business documents

RAG pipeline

```
Data Source
↓
Text Extraction
↓
Cleaning
↓
Chunking
↓
Embeddings
↓
PostgreSQL + pgvector
↓
Similarity Search
↓
```

Relevant Context

↓

LLM

↓

Grounded Response

By retrieving the most relevant chunks of verified business content and supplying them to the LLM as context before generation, the system substantially reduces the likelihood of the model fabricating (hallucinating) information. The model is instructed to answer from the retrieved context and to acknowledge when information is not available, rather than guessing. When a customer's own image or document is analyzed (Section 7), its extracted content is merged into this same context window alongside the retrieved knowledge.

Unanswered Questions feature

Customer asks a question (text, voice, or on a call)

↓

Knowledge retrieval fails or confidence is low

↓

Question is logged

↓

Administrator reviews it

↓

Answer / knowledge is added

↓

Future queries can use the new information

7. Voice & Media Intelligence

This layer extends the platform beyond plain text so the AI can understand everything a customer naturally sends through WhatsApp: a spoken voice note, a photo, or a scanned document.

7.1 Voice Message Pipeline

```
Customer sends a WhatsApp voice note
↓
WhatsApp Business Cloud API delivers the media
↓
Audio is downloaded
↓
Speech-to-Text transcription
↓
Transcript enters the normal agent pipeline (Supervisor → RAG / Lead / Tools)
↓
Text response generated
↓
(Optional) Text-to-Speech converts the reply back into a voice note
↓
Response delivered to the customer on WhatsApp
```

Because the transcript is treated exactly like a typed message, no part of the Supervisor Agent, RAG pipeline, or lead-qualification logic needs to change to support voice — it is a normalization step at the front of the pipeline.

7.2 Image & Document Understanding

```
Customer sends an image or document (photo, screenshot, PDF, scanned form)
↓
Media is downloaded and stored
↓
Vision-capable model analyzes the content
↓
Extracted description / structured data is produced
↓
Result is merged into the conversation context
↓
Supervisor Agent uses it for intent detection, support, or lead extraction
```

Representative use cases

- A customer photographs a damaged or faulty product for a support claim
- A customer sends a screenshot of an error message or a failed payment
- A customer shares a business card or ID document for verification
- A customer fills out and sends a scanned requirement form or quotation PDF
- A customer sends a photo of a competitor's product for a comparison request

7.3 Data Flow Summary

Media Type	Processing Step	Output Fed to Supervisor
Voice note	Speech-to-Text (transcription)	Plain-text transcript
Image / photo	Vision model analysis	Structured description of image contents
Document (PDF, scan)	Vision / OCR + document analysis	Extracted text and key fields
Outgoing reply (optional)	Text-to-Speech synthesis	Voice note sent back to customer

8. AI Voice Call Agent

The platform can now answer a real phone call — the AI listens to the caller and answers back, live, in a natural spoken conversation, rather than only handling asynchronous WhatsApp messages.

8.1 How It Works

```
Incoming phone call
↓
Telephony Provider (e.g., Twilio Voice)
↓
Real-time audio stream (WebSocket)
↓
Streaming Speech-to-Text
↓
Voice Call Agent → Supervisor Agent (same routing as WhatsApp)
↓
Knowledge Agent / Lead Agent / Scheduling Agent, as needed
↓
Response text generated
↓
Streaming Text-to-Speech
↓
Audio played back to the caller
```

8.2 Capabilities

- **AI listens to and answers the call** — the caller speaks naturally, without pressing keys or following an IVR menu
- Answers FAQs and business questions from the same RAG-grounded knowledge base used on WhatsApp
- Qualifies the caller into a structured lead, exactly as a WhatsApp conversation would, and syncs it to HubSpot
- Books a meeting directly on the call using the Scheduling Agent
- Detects when it should not continue and transfers the live call to a human team member
- Handles interruptions / barge-in so the caller can speak over the AI naturally
- Works around the clock, so no inbound call is missed outside business hours

8.3 Call Records & Auditability

Every call is recorded and transcribed, and the resulting transcript, summary, extracted lead, and any calendar booking are stored against the same customer profile as their WhatsApp history, visible in the React dashboard alongside chat conversations (Section 14).

8.4 Compliance Note

Call recording and AI-handled calls are subject to regional consent and telecom-compliance requirements (e.g., call-recording disclosure, data-retention rules). The architecture logs consent status per call and keeps recordings inside the workspace-isolated data boundary described in Section 17.

9. Frappe CRM & ERPNext Integration

Many businesses already run their accounting, customers, employees, and sales pipeline on their own servers. This revision connects the platform to the company's **self-hosted ERPNext** and **Frappe CRM**, so the AI works with the business's real records: it captures leads into the CRM and, after identity verification, serves employee and customer information in the WhatsApp chat — while the data stays on the company's own infrastructure.

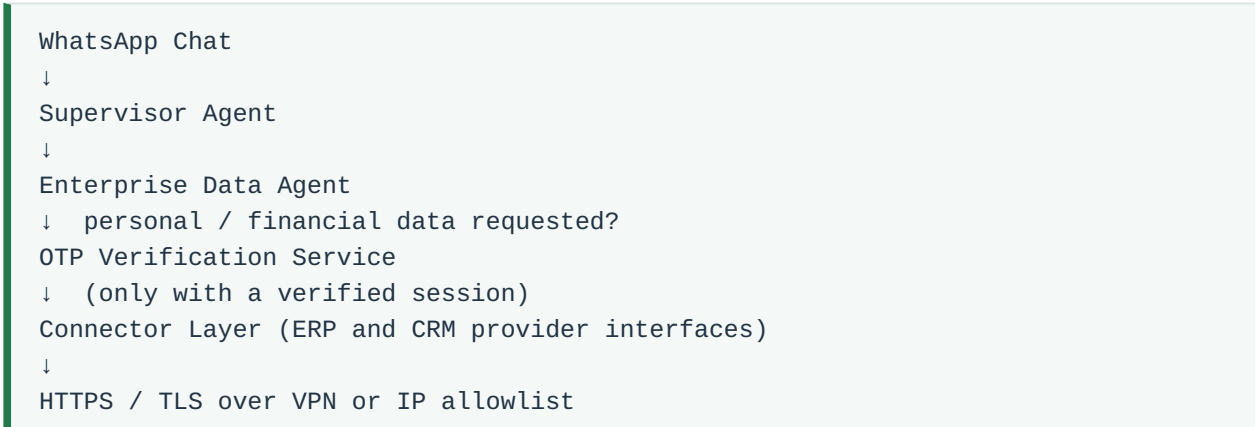
9.1 The Frappe Ecosystem at a Glance

- **Frappe Framework** — the open-source, full-stack Python/JavaScript framework underneath both products. Data is stored as **DocTypes** (record types such as Employee or Sales Invoice), and every DocType automatically gets a REST API, role-based permissions, and webhook support.
- **ERPNext** — open-source ERP built on Frappe, covering accounting and invoicing, customers, sales and purchasing, inventory, and more. It can be self-hosted on the company's own server.
- **Frappe HR (HRMS)** — the separate HR and payroll app that runs alongside ERPNext (leave, attendance, payslips, expense claims). Leave, attendance, and payslip lookups need it to be installed.
- **Frappe CRM** — open-source CRM built on Frappe for managing leads, deals, contacts, organizations, notes, and tasks, designed to work alongside ERPNext.

Product	Role in this platform	Key DocTypes used
ERPNext	System of record for customers, invoices, payments, and the employee master	Employee, Customer, Contact, Sales Invoice, Payment Entry, Delivery Note
Frappe HR (optional)	Leave, attendance, and payroll data for employee self-service	Leave Allocation, Leave Application, Attendance, Salary Slip
Frappe CRM	Destination for AI-captured leads and deals	CRM Lead, CRM Deal, CRM Organization, Contact

DocType and field names shown here are the standard ones. They can differ between versions and after customization, so the connector reads them from a per-workspace mapping and allowlist, and each mapping should be validated against the installed version before go-live.

9.2 Integration Architecture





- **Protocol** — Frappe's REST API: `/api/resource/<DocType>` to read and filter records and `/api/method/<path>` for whitelisted server methods.
- **Authentication** — token authentication with an API key and secret generated for a dedicated integration user (Authorization: token api_key:api_secret), held in the secret manager and never in source control or logs.
- **Least privilege** — the integration user gets a custom role with read-only permission on the allowlisted DocTypes, and create/write only on CRM Lead and CRM Deal. Frappe's own role and field-level permissions are the first barrier; the platform's DocType/field allowlist is the second.
- **Network** — the server is reached over TLS through a VPN, private network, or IP allowlist. The ERP database is never exposed.
- **Resilience** — timeouts, retries with backoff, and a circuit breaker. If the server is unreachable the AI says the information is temporarily unavailable and offers a human; it never guesses values.
- **No lingering copies** — personal data is not cached beyond the verified session and is never written into the RAG index or long-term memory.
- **Multi-tenant** — each workspace has its own connection record, so one workspace can never reach another's server.

9.3 Lookup-to-API Mapping

Lookup	Source	Illustrative call
Employee identity check and profile	ERPNext — Employee	GET <code>/api/resource/Employee/<id></code> with a fixed fields list (e.g., employee_name, designation, department, branch, date_of_joining, reports_to, status, cell_number)
Leave balance / recent leave	Frappe HR — Leave Allocation, Leave Application	GET <code>/api/resource/Leave Application</code> filtered by the verified employee
Attendance summary	Frappe HR — Attendance	GET <code>/api/resource/Attendance</code> filtered by employee and date range
Invoice and customer number match	ERPNext — Sales Invoice, Customer, Contact	GET <code>/api/resource/Sales Invoice/<id></code> (invoice number, customer, contact_mobile), cross-checked with the customer's Contact mobile
Payments against an invoice	ERPNext — Payment Entry	Payment Entry references filtered by the invoice
Delivery status	ERPNext — Delivery Note	Delivery Note linked to the invoice / sales order
Create a lead	Frappe CRM — CRM Lead	POST <code>/api/resource/CRM Lead</code>

9.4 Pluggable CRM: Frappe CRM or HubSpot

Each workspace chooses where qualified leads go: self-hosted Frappe CRM, HubSpot, or both. The agents never call a CRM directly; they call one provider-agnostic tool and a provider interface does the rest.

```
Lead Qualification Agent
↓
sync_lead_to_crm()
↓
CRM Provider Interface
↓
FrappeCRMProvider      HubSpotProvider
```

Aspect	Frappe CRM (self-hosted)	HubSpot (cloud)
Hosting	On the company's own server; data stays in-house	HubSpot-hosted SaaS
Authentication	API key and secret for a dedicated integration user	Scoped OAuth 2.0
Lead objects	CRM Lead, convertible to CRM Deal	Contact and Deal
ERP linkage	Built to work alongside ERPNext (same customers and contacts)	None natively; leads sync one way
Licensing	Open source; self-hosting cost only	Commercial, tiered subscription
Best when	The company already runs ERPNext or wants data sovereignty	The company already uses HubSpot sales or marketing tools

Employee and invoice look-ups always use ERPNext, whichever CRM is selected. If both CRMs are enabled, leads are written to both (dual-write) and each destination's external ID is stored so records are updated rather than duplicated.

Default lead field mapping (editable per workspace)

Platform field	Frappe CRM	HubSpot
name	first_name / last_name	firstname / lastname
email	email	email
phone	mobile_no	phone
company	organization	company
requirement / service	notes or custom field	deal description or custom property
budget	deal value or custom field	deal amount
timeline, lead score	custom fields	custom properties
source	source (WhatsApp / Phone call)	lead source property
conversation or call summary	note / comment on the lead	note on the contact or deal

10. OTP-Verified Self-Service on WhatsApp

Employees and customers can retrieve their own information from ERPNext by chatting on WhatsApp. Because this is personal and financial data, nothing is shown until the requester has been matched to the ERP record **and** has confirmed a one-time password (OTP) sent on WhatsApp.

10.1 Employee Self-Service

```
Employee sends Employee ID (e.g., HR-EMP-00042)
↓
Enterprise Data Agent → OTP Verification Service
↓
ERPNext check: employee exists, status Active, registered mobile number
↓
Registered number = WhatsApp sender number?
  no → generic "we could not verify these details" reply
  ↓ yes
OTP sent on WhatsApp (authentication template)
↓
Employee replies with the OTP (5-minute expiry, max 3 attempts)
↓
Verified session opened (15 minutes, bound to this chat and number)
↓
Details fetched live from ERPNext and shown in the chat
```

Within the verified session the employee can ask follow-ups such as "what is my leave balance?". Every answer is limited to that employee's own record.

10.2 Customer Invoice Lookup

```
Customer sends Invoice ID and their WhatsApp number
↓
Verification Service: number typed = number this chat is coming from?
↓
ERPNext check: invoice exists and its contact / customer mobile = that number?
  no → generic "we could not verify these details" reply
  ↓ yes
OTP sent on WhatsApp to that number
↓
Customer replies with the OTP
↓
Verified session opened (bound to this number and this invoice's customer)
↓
Invoice details fetched live from ERPNext and shown in the chat
```

The customer sees only invoices that belong to the customer record tied to their verified number. Asking for another customer's invoice returns the same generic reply as an invalid ID.

10.3 OTP Design

Parameter	Design
Generation	6-digit numeric code from a cryptographically secure random source
Delivery	WhatsApp Business Cloud API, authentication-category message template
Storage	Only a keyed hash (HMAC) in Redis with a time-to-live; the plain code is never stored or logged
Expiry	5 minutes; single use, consumed on success
Wrong attempts	Maximum 3, then the challenge is invalidated
Resend limits	60-second cool-down; maximum 3 OTPs per number per hour
Lockout	30-minute lock after repeated failures, with an admin alert on suspicious patterns
Comparison	Constant-time comparison
Binding	Challenge tied to the phone number, conversation, purpose, and subject (employee ID or invoice ID)
Verified session	15-minute lifetime with idle timeout; revocable by admin or on human takeover

Design note. The OTP is delivered to the same WhatsApp number the person is chatting from, so on its own it does not prove who is holding the phone. The identity assurance comes from the combination: the WhatsApp number must match the number registered in ERPNext, and the requester must prove live control of it. For higher-sensitivity data, a step-up check (for example an OTP to the company email, or dashboard approval) is recommended.

10.4 What Can Be Shown

Field visibility is configured per workspace by an administrator and enforced as an allowlist.

Tier	Employee data (ERPNext / Frappe HR)	Customer data (ERPNext)	Access rule
1 — Basic	Name, designation, department, branch, date of joining, reporting manager, official email, employment status	Invoice number, date, due date, status, customer name	OTP-verified session
2 — Operational	Leave balance and recent leave, attendance summary	Line items, taxes, grand total, amount paid, outstanding amount, payment history, delivery status	OTP-verified session

3 — Sensitive	Payslips and salary, bank and government ID details	Not offered by default	Disabled by default; if enabled, needs a fresh OTP (step-up) and masked account and ID numbers
Never	Other people's records, internal cost or margin, credentials, full bank or ID numbers, internal free-text notes		Blocked by the field allowlist

10.5 Security and Privacy Controls

- **Deterministic verification** — identity checks run in application code, not in the LLM; the model cannot skip, forge, or "decide" verification.
- **Identity-bound tools** — the subject of every lookup comes from the server-side session, never from model output or the user's message.
- **Anti-enumeration** — an unknown ID, an inactive employee, a number mismatch, and a wrong customer all produce the same generic reply, with timing kept similar.
- **Rate limiting** — per number, per ID, and per IP, on both lookups and OTP requests.
- **Data minimisation** — only allowlisted fields are fetched; the LLM sees them only for the current verified subject.
- **Masking and redaction** — sensitive values are masked or redacted in stored transcripts and logs; OTP values are never stored.
- **Prompt-injection safety** — text returned from ERP or CRM fields is treated as data, never as instructions.
- **Audit trail** — every OTP challenge, verification result, and ERP fetch (which fields, for which subject) is logged and visible to admins, without the data itself.
- **Privacy compliance** — employee and customer notice, purpose limitation, and retention rules aligned with applicable data-protection law (for example India's DPDP Act, 2023).

11. Unique & Future-Proof Capabilities

NEW SECTION

This section covers five advanced capabilities added in this revision: a dynamic, headless-CMS-backed knowledge base; marketing-cloud syncing; AI-initiated outbound campaigns; live sentiment analysis with auto-escalation; and controlled ERP write actions. Together they move the platform from a system that answers and listens to one that also keeps its own knowledge current, reaches out proactively, senses distress, and can act on a verified employee's behalf.

11.1 Strapi Headless CMS — Dynamic Knowledge Base

Why: static PDF uploads go stale the moment a price or FAQ changes. A headless CMS lets a non-technical team keep the content the AI relies on current, structured, and versioned.



- **Content types** — Products, Pricing Plans, FAQs, Service Descriptions, Promotions, modeled as Strapi collection types
- **Sync trigger** — Strapi's publish/unpublish webhook re-indexes only the changed entry, so updates typically reach the AI within minutes
- **Draft safety** — only entries in the Published state are ever ingested; Draft and unpublished content is invisible to the AI and to customers
- **Coexistence** — Strapi becomes one more knowledge source alongside PDFs, URLs, and manual text (Section 6); it does not replace them
- **Authentication** — a scoped Strapi API token restricted to read access on the relevant content types

Aspect	Static PDF Knowledge Source	Strapi-Backed Knowledge Source
Update process	Re-export and re-upload a document	Edit and publish an entry in a content editor UI
Time to reflect a change	Manual, whenever someone re-uploads	Minutes, via webhook-triggered re-index
Structure	Unstructured text, chunked generically	Structured fields (price, SKU, category) usable for precise answers
Ownership	Usually IT or the AI team	Marketing/product team, no engineering involvement

11.2 Adobe Experience Cloud Syncing

Why: the sales CRM (Frappe CRM / HubSpot, Section 9.4) tracks the pipeline; the marketing stack needs the same signal to personalize email and ad retargeting.

```
Qualified lead or engaged conversation
↓
Lead Qualification Agent (as in Section 5)
↓
sync_lead_to_crm()          sync_to_adobe_experience_cloud()
↓                            ↓
Frappe CRM / HubSpot        Adobe Marketo Engage or Adobe Campaign
(pipeline / deals)          (nurture journeys, ad retargeting)
```

- **Data pushed** — contact details (where consented), lead source, qualification stage, topics/intent discussed, and engagement recency
- **Destination choice** — Marketo Engage for B2B lead nurturing and scoring, or Adobe Campaign for broader cross-channel (email/SMS/ad) journeys; configurable per workspace
- **Consent-aware** — only synced for contacts who have opted in to marketing communication, tracked separately from the WhatsApp service opt-in
- **Authentication** — Adobe's OAuth Server-to-Server credentials, scoped to the specific Experience Cloud instance and API

11.3 Proactive Outbound Campaigns

Why: this promotes the previously future-scoped "outbound AI-initiated calls and messages" into the active architecture, driven by the Outbound Campaign Agent (Section 4.E, 5).

```
Trigger source: schedule (reminder) · ERPNext event (invoice overdue, leave
approved) · CRM stage change
↓
Outbound Campaign Agent
↓
Opt-in check + quiet-hours / messaging-window check
↓
WhatsApp template message      or      Outbound call via Voice Call Agent
↓
Customer/employee reply re-enters the normal Supervisor Agent pipeline
```

Example triggers

- Appointment / meeting reminder ahead of a calendar booking (Section 8)
- Invoice follow-up when an ERPNext Sales Invoice becomes overdue
- ERP-triggered notice — e.g., a submitted Leave Application is approved (Section 11.5), a Delivery Note is dispatched
- **Compliance** — every outbound send checks WhatsApp opt-in status and uses an approved message template category; outbound calls follow the same call-recording consent rules as inbound (Section 8.4)
- **Rate control** — per-contact frequency caps prevent repeated or duplicate outreach for the same event

-
- **Auditability** — every outbound send is logged with its trigger, template, and outcome

11.4 Live Sentiment Analysis on Calls

Why: a frustrated caller should reach a human before the AI's confidence-based escalation alone would trigger it.

```
Live call audio + streaming transcript (Section 8.1)
↓
Sentiment Monitoring Agent scores each turn
↓
Sustained negative trend detected
↓
Human Escalation Agent notified
↓
Call is transferred to a human, exactly as in Section 8.2 / Section 13
```

- **Signal sources** — transcript-based sentiment (word choice, negation, complaint language) combined with voice-based signal (pace, volume, tone) where the telephony/STT provider exposes it
- **Trend, not single turn** — escalation triggers on a sustained negative trend across several turns, not one frustrated sentence, to avoid over-triggering
- **Transparent handoff** — the caller is told a team member is joining, and the human sees the sentiment trend and transcript so far, the same way a low-confidence escalation is presented
- **Feeds analytics** — sentiment trend is stored per call for reporting (Section 16) without needing separate infrastructure

11.5 ERP Write Actions — Employee Self-Service, Beyond Read-Only

Why: Section 10's employee self-service is read-only by design. This capability extends it to a small, explicit set of write actions, without weakening the identity-verification model.

```
Verified employee session already open (Section 10.1 – OTP required first)
↓
Employee: "I want to apply for 2 days leave from 14 to 15 Oct"
↓
Enterprise Data Agent drafts the ERPNext record (Leave Application / Expense Claim / HD Ticket)
↓
Plain-language summary shown back to the employee for confirmation
↓
Employee confirms ("yes" / "confirm")
↓
submit_leave_application() writes to ERPNext, status: Pending Approval
↓
Normal ERPNext approval workflow proceeds (manager notified, as usual)
↓
Approval/rejection can trigger an outbound WhatsApp notice (Section 11.3)
```

Supported write actions

Action	ERPNext / Frappe HR DocType	Resulting state
Apply for leave	Leave Application	Created as Pending Approval — never auto-approved by the AI
Raise an expense claim	Expense Claim	Created as Draft/Pending Approval, with receipt image attached if provided (Section 7.2)
Open a support/IT ticket	HD Ticket (Frappe Helpdesk) or a configured Issue DocType	Created and routed to the relevant team's normal queue

Guardrails

- **Verified session required** — no write action is available before the OTP verification in Section 10 succeeds
- **Confirm-then-submit** — the AI never submits on the first message; it restates what it is about to create and waits for explicit confirmation
- **No approval bypass** — writes land in the same Pending/Draft state and approval workflow a human-submitted request would, so existing manager approvals still apply
- **Scoped write permission** — the integration user's write access is limited to exactly these DocTypes, separate from its broader read allowlist (Section 9.2)
- **Full audit trail** — every write action is logged with the verified employee, the exact payload submitted, and the ERPNext response
- **Reversible by design** — because actions are created in a pending state, a human can review, edit, or reject them through ERPNext's normal UI before anything takes effect

12. Business Persona Configuration

Each business owner can customize how their AI agent behaves, so the assistant reflects their brand and communication style rather than sounding generic — across chat, voice replies, and phone calls.

Configurable fields

- AI agent name
- Business name
- Business description
- Agent role
- Tone
- Response length
- Language
- Emoji behavior
- Voice persona (for voice-note replies and phone calls)
- Business goals
- Custom instructions
- Restrictions
- Escalation rules

Example tones

- Professional
- Friendly
- Casual
- Concise

Language capabilities

- English
- Hindi
- Marathi
- Hinglish
- Mixed-language conversations within a single chat or call

13. Lead Generation and Qualification

The platform converts ordinary WhatsApp conversations — and now phone calls — into structured, actionable leads by progressively collecting the information a sales team needs.

Example — Customer: "I need a mobile application for my business."

Name
↓
Business
↓
Requirement

↓
Budget
↓
Timeline
↓
Contact Information

...and then creates a structured lead record. Example:

```
{
  "name": "Rahul",
  "service": "Mobile App Development",
  "requirement": "E-commerce mobile application",
  "budget": "₹3-5 lakh",
  "timeline": "2 months",
  "source": "WhatsApp",
  "lead_status": "qualified"
}
```

Once captured, leads are automatically created or updated in the workspace's chosen CRM **UPDATED** — as Leads and Deals in self-hosted **Frappe CRM**, as Contacts and Deals in **HubSpot**, or in both — keeping the sales pipeline synchronized in real time with no manual data entry, and no lead lost between a chat, a voice note, or a phone call. Provider selection and field mapping are described in Section 9.4. Leads can additionally be sent to other external systems via outbound webhooks or API calls where needed.

14. Human Takeover

Automation is powerful but should never fully replace human judgment for sensitive, high-value, or ambiguous conversations. The dashboard provides a complete human-takeover workflow across both chat and voice.

Features

- Live conversation and live call monitoring
- Full visibility into the AI's chat history and call transcripts
- One-click human takeover of a chat, or live transfer of a phone call
- Automatic AI pause once a human takes over
- Human response directly from the dashboard
- Conversation or call assignment to a specific team member
- Ability to resume AI handling after the human interaction concludes

Why this matters

For real business usage, blind automation is a liability: mishandled complaints, sensitive negotiations, or edge cases the AI was not designed for can damage customer trust if left entirely to the model. Human takeover ensures the business always retains control, while still benefiting from automation for the majority of routine conversations and calls.

15. React Dashboard

The dashboard is the operational control center for the business owner and team — it surfaces everything the AI is doing and lets humans intervene where needed.

Section	Purpose
Dashboard	High-level overview: key metrics, recent activity, system health at a glance.
WhatsApp	Manage the connected WhatsApp Business account, templates, and connection status.
Inbox	Unified inbox for incoming customer messages requiring attention.
Conversations	Full conversation history and thread view for every customer, including transcribed voice notes and analyzed images/documents.
Calls	Live and historical phone call monitoring, recordings, transcripts, and one-click call transfer to a human.
AI Agent	Configure agent persona, tone, language, voice, and behavior rules.
Knowledge Base	Upload and manage PDFs, URLs, FAQs, and manual knowledge entries.
Leads	View, filter, and manage structured leads captured by the AI, with CRM sync status (Frappe CRM / HubSpot). UPDATED
Playground	Sandbox environment for testing prompts, persona changes, and agent behavior.
Analytics	Reporting on conversations, calls, leads, escalations, and AI performance.
Integrations UPDATED	Manage the ERPNext / Frappe CRM server connection (URL, API credentials, DocType and field allowlist), HubSpot connection, CRM provider selection, calendar connection, and webhooks.
Verification Logs	Audit view of OTP challenges, verification outcomes, lockouts, and ERP lookups (never shows OTP values or the fetched data).
Knowledge Base UPDATED	Upload and manage PDFs, URLs, FAQs, manual entries, and connect Strapi as a live content source.
Campaigns NEW	Configure and monitor proactive outbound WhatsApp/voice campaigns and their triggers.
Employee Requests NEW	Review, approve, or reject ERP write actions (leave, expense claims, tickets) raised via WhatsApp, alongside ERPNext's own approval screens.
Settings	Workspace, user, security, and account-level configuration.

16. AI Playground

The Playground is a sandbox environment that lets a business owner or developer safely test and iterate on the AI agent without affecting live customer conversations or calls.

What can be tested

- AI responses to sample messages, voice notes, images, and documents

-
- Simulated phone-call conversations
 - Prompt changes
 - Persona changes (tone, language, voice, restrictions)
 - Knowledge retrieval behavior
 - RAG responses and grounding quality
 - Agent routing decisions
 - Tool usage, including calendar booking, CRM sync (Frappe CRM / HubSpot), and ERP look-ups against a test ERPNext site with simulated OTP **UPDATED**

Debugging information displayed

- Selected agent
- Retrieved knowledge (which chunks/sources were used)
- Media/voice transcription output, where applicable
- Tool calls made during the exchange
- Confidence score
- Final response

17. Analytics and Monitoring

The platform tracks operational and AI-quality metrics to give the business full visibility into system performance.

Planned metrics

- Total conversations and calls
- AI-resolved conversations / calls
- Human escalations and call transfers
- Leads generated / qualified leads / CRM sync success rate (Frappe CRM / HubSpot)
- Calendar bookings made
- Verification requests, OTP success and failure rates, lockouts
- Employee and customer self-service look-ups completed
- ERP / CRM API latency and error rate
- Strapi content sync success rate and freshness **NEW**
- Leads/records synced to Adobe Experience Cloud **NEW**
- Outbound campaigns sent, delivered, and responded to **NEW**
- Calls escalated due to negative sentiment **NEW**
- ERP write actions submitted, approved, and rejected **NEW**
- Response time (chat) and average handle time (calls)
- Voice note transcription volume and accuracy
- Image/document analysis volume
- Unanswered questions
- Agent executions
- Tool executions
- RAG performance
- Error rates

-
- Token usage / minutes of voice processed

Observability

Observability is implemented using structured logs and OpenTelemetry, allowing traces to be followed end-to-end — from an incoming WhatsApp message or phone call, through agent routing, media analysis, and tool calls, to the final outgoing response — for debugging and performance analysis.

18. Security and Governance

Because the platform executes real actions on behalf of a business (sending messages, answering calls, storing customer data, calling external systems), security and governance are treated as first-class architectural concerns rather than an afterthought.

- JWT authentication
- Role-Based Access Control (RBAC)
- Workspace isolation (multi-tenant data separation)
- Input validation on all API and webhook payloads
- Webhook signature verification
- API authentication for all backend endpoints
- Secret management (API keys, tokens kept out of source control)
- Rate limiting
- Audit logging of key actions and decisions, including tool calls made during a live call
- PII protection for customer data
- Secure storage of call recordings and uploaded media, with access controls
- Call-recording consent tracking
- Tool permission control — agents can only call tools they are authorized to use
- Prompt injection protection
- RAG data isolation between workspaces
- Least-privilege tool execution
- Scoped OAuth credentials for HubSpot and calendar connections
- OTP verification before any ERP or CRM personal or financial data is shared
- Least-privilege Frappe / ERPNext integration user (read-only, DocType and field allowlist)
- Identity-bound tools: the subject comes from the server-side verified session, never from model output
- Anti-enumeration replies, OTP attempt and resend limits, lockouts
- Encrypted, network-restricted connectivity (TLS, VPN / IP allowlist) to the self-hosted ERP
- Masking and redaction of sensitive ERP data in transcripts and logs
- Scoped, separate write-permission role for ERP write actions, distinct from the read allowlist **NEW**
- Confirm-then-submit flow — no ERP write happens on a single unconfirmed message **NEW**
- Writes always land in a Pending/Draft approval state — no AI auto-approval **NEW**
- Consent-gated marketing sync to Adobe Experience Cloud, separate from WhatsApp service opt-in **NEW**
- Opt-in and quiet-hours enforcement on all proactive outbound campaigns **NEW**

A core governance principle of this system: LLMs are never given unrestricted access to tools. Every tool call — including booking a calendar event, syncing to a CRM, fetching an ERP record, or transferring a live call — is checked, validated, and logged before it is allowed to execute.



```
graph TD; A[LLM requests tool] --> B[Permission Check]; B --> C[Input Validation]; C --> D[Risk / Policy Check]; D --> E[Tool Execution]; E --> F[Audit Log];
```

LLM requests tool
↓
Permission Check
↓
Input Validation
↓
Risk / Policy Check
↓
Tool Execution
↓
Audit Log

19. Technology Stack

Frontend

Layer	Technology
UI Library	React
Language	TypeScript
Build Tool	Vite
Styling	Tailwind CSS
Routing	React Router
Server State	TanStack Query
Client State	Zustand

Backend

Layer	Technology
Language	Python
Web Framework	FastAPI
Data Validation	Pydantic
ORM	SQLAlchemy
Migrations	Alembic

AI

Component	Technology
LLM Provider	OpenAI API
Agent Orchestration	LangGraph
Composability Utilities	LangChain (where useful)
Embeddings	OpenAI Embeddings
Vision / Image & Document Understanding	OpenAI multimodal (vision) model
Speech-to-Text	OpenAI Whisper API (streaming variant for live calls)
Text-to-Speech	OpenAI TTS API
Action Execution	Tool Calling
Structured Data	Structured Outputs

RAG

Component	Technology
Database	PostgreSQL
Vector Search	pgvector
PDF Parsing	PyMuPDF
Web Extraction	BeautifulSoup / Crawl4AI

WhatsApp & Voice Telephony

Component	Technology
Messaging Platform	WhatsApp Business Cloud API
Event Delivery	Meta Webhooks
Voice Calling / Telephony	Twilio Voice API (real-time media streaming)
Call Audio Transport	WebSocket-based bidirectional audio stream

ERP, CRM & Calendar Integrations UPDATED

Component	Technology
ERP (self-hosted)	ERPNext via the Frappe REST API (/api/resource, /api/method)
HR data (optional)	Frappe HR (HRMS) DocTypes: leave, attendance, payroll
CRM option 1 (self-hosted)	Frappe CRM (CRM Lead, CRM Deal) via the Frappe REST API
CRM option 2 (cloud)	HubSpot API (Contacts, Deals, Timeline events)
Connector pattern	Provider interface (adapter pattern) with an async httpx client, timeouts, retries, circuit breaker
Frappe authentication	API key and secret token for a dedicated integration user
Private connectivity	TLS over VPN / private network / IP allowlist
Headless CMS NEW	Strapi (REST / GraphQL API, publish webhooks)
Marketing Cloud NEW	Adobe Marketo Engage / Adobe Campaign (Adobe Experience Cloud APIs)
Outbound Campaigns NEW	WhatsApp template messages + outbound Voice Call Agent, scheduled via background worker
Live Sentiment Analysis NEW	Streaming transcript sentiment scoring alongside the STT pipeline (Section 8)
ERP Write Actions NEW	Frappe REST API POST/PUT with a scoped write-permission role, confirm-then-submit flow
Calendar	Google Calendar API / Microsoft Outlook (Graph API)

Async Processing & Real-Time

Component	Technology
Queue / Cache	Redis
Background Processing	Background Worker
Real-Time Updates	WebSockets

Security

Component	Technology
Auth Tokens	JWT
Access Control	RBAC
Credential Storage	Password hashing
Webhook Integrity	Webhook verification
Third-Party OAuth	Scoped OAuth 2.0 for HubSpot / calendar providers
OTP Verification	Secure random generation, HMAC-hashed codes in Redis with TTL, WhatsApp authentication templates
Adobe Authentication NEW	Adobe OAuth Server-to-Server credentials, scoped per Experience Cloud product

Testing & Code Quality

Component	Technology
Backend Testing	pytest, httpx
Frontend Testing	React Testing Library, Playwright
Python Linting	Ruff
Python Type Checking	MyPy
JS/TS Linting	ESLint
Formatting	Prettier

DevOps & Observability

Component	Technology
Containerization	Docker
Local Orchestration	Docker Compose
Version Control	Git
Hosting / CI	GitHub
Tracing	OpenTelemetry

Logging	Structured Logging
---------	--------------------

20. Complete System Architecture

The core message-processing pipeline flows as follows:

```
Customer
↓
WhatsApp
↓
WhatsApp Business Cloud API
↓
FastAPI
↓
Webhook / Message Processor (text / voice note / image / document)
↓
Media Understanding Agent (transcribe / analyze, when needed)
↓
Redis Queue
↓
Agent Workflow
↓
Supervisor Agent
↓
Specialized Agents
↓
RAG / Tools / ERPNext (read & approved writes) / CRM (Frappe CRM or HubSpot) /
Strapi / Adobe Experience Cloud / Calendar / Database
↓
Response Validation
↓
WhatsApp
↓
Customer
```

In parallel, live phone calls flow through a dedicated real-time path:

```
Caller
↓
Telephony Provider (Twilio Voice)
↓
WebSocket Audio Stream
↓
Streaming Speech-to-Text
↓
Voice Call Agent → Supervisor Agent (same specialized agents / RAG / tools as
above)
↓
Streaming Text-to-Speech
↓
Caller
```

Verified ERP look-ups and CRM sync pass through a dedicated connector layer to systems the company hosts itself:

```
Enterprise Data Agent
↓
OTP Verification Service (Redis: hashed OTP, verified session)
↓
Connector Layer: ERP provider · CRM provider interface
↓
Frappe REST API over TLS / VPN
↓
Company-hosted ERPNext · Frappe CRM (HubSpot: cloud, via OAuth)
```

NEW Knowledge and marketing data flow through their own integrations alongside the ERP/CRM connector layer described in Section 9.2:

```
Strapi CMS (publish webhook) → RAG ingestion (Section 6 / 11.1)
Lead Qualification Agent → Adobe Experience Cloud API (Section 11.2)
Outbound Campaign Agent → WhatsApp / Voice Call Agent (Section 11.3)
```

Dashboard connectivity: the React Dashboard communicates with the backend in real time.

```
React Dashboard
↓
FastAPI
↓
WebSocket
```

Data layer: PostgreSQL with the pgvector extension serves as the system of record for both relational and vector data, connected to the following core entities:

Users · Workspaces · Conversations · Messages · Calls · Media Assets · Agents · Knowledge · Leads · CRM Syncs · CRM / ERP Connections · OTP Challenges · Verified Sessions · Calendar Bookings · Strapi Knowledge Sources · Marketing Cloud Syncs · Outbound Campaigns · Call Sentiment Scores · ERP Write Requests · Tools · Audit Logs

This diagram reflects the current/planned architecture as described throughout this document. Components marked elsewhere as Planned or Future Scope are not part of the implemented pipeline yet.

21. End-to-End Examples

20.1 WhatsApp Text/Lead Example

Customer: "I want to build an ecommerce website for my clothing business. How much will it cost?"

1. WhatsApp receives the message.
2. Meta sends a webhook event to the platform.
3. FastAPI validates the webhook signature.
4. The message is stored in the database.
5. The message enters the processing queue (Redis).
6. The Supervisor Agent analyzes intent.
7. The Supervisor identifies a service inquiry with lead potential.
8. The Knowledge Agent retrieves relevant business information via RAG.
9. The Sales / Lead Agent continues the qualification conversation.
10. The customer provides budget and timeline.
11. The Lead Qualification Agent extracts structured lead data.
12. The lead is stored in the database and synced to the configured CRM (Frappe CRM or HubSpot).
13. The AI sends its response to the customer over WhatsApp.
14. The conversation becomes visible in the React dashboard in real time.
15. A human can take over the conversation if necessary.
16. The complete workflow — every step, agent decision, and tool call — is logged.

20.2 Voice Note + Document Example

Customer sends a voice note: "I'm sending you our current logo, can you quote a redesign?" followed by an image of the logo.

1. WhatsApp receives the voice note and the image.
2. The Media Understanding Agent transcribes the voice note to text and analyzes the image.
3. Both results are merged into the conversation context.
4. The Supervisor Agent identifies a design-service inquiry with lead potential.
5. The Knowledge Agent retrieves relevant pricing/service information via RAG.
6. The Lead Qualification Agent captures requirement, budget, and timeline over the conversation.
7. The lead is stored and synced to the configured CRM as a new Lead/Deal.
8. The Scheduling Agent offers available slots and books a call once the customer agrees.
9. A confirmation is sent back to the customer over WhatsApp.

20.3 Inbound Phone Call Example

A prospective customer calls the business number outside office hours.

1. The call reaches the Telephony Provider and is routed to the AI Voice Call Agent.
2. The AI answers and greets the caller using the configured voice persona.
3. The caller explains they need a mobile app; the AI listens via streaming speech-to-text.
4. The transcript is routed to the Supervisor Agent, which engages the Knowledge and Lead Agents exactly as it would on WhatsApp.

5. The AI answers pricing questions from the RAG knowledge base, speaking the response back via text-to-speech.
6. The Lead Qualification Agent captures the caller's requirement, budget, and timeline during the conversation.
7. The AI offers to book a follow-up call, checks calendar availability, and books the meeting.
8. The lead and the call recording/transcript are stored and synced to the configured CRM.
9. A WhatsApp confirmation message is sent to the caller's number with the booking details.
10. The call appears in the dashboard's Calls section for review; a human could have taken it over live if needed.

20.4 Employee Self-Service Example

Illustrative chat (sample data):

```
Employee: HR-EMP-00042
AI:       Thanks. For your security I've sent a 6-digit code to your
         registered WhatsApp number. Please reply with it (valid 5 min).
Employee: 482913
AI:       Verified. Here are your details:
         Name: Priya Sharma · Designation: Sales Executive
         Department: Sales · Reports to: A. Kulkarni
         Date of joining: 12 Jan 2023 · Status: Active
Employee: How many leaves do I have left?
AI:       Casual leave: 6 days · Sick leave: 4 days
```

1. The Supervisor detects an employee self-service intent and routes to the Enterprise Data Agent.
2. The Verification Service fetches the Employee record from ERPNext and compares its registered mobile with the WhatsApp sender number.
3. The numbers match, so an OTP is sent through a WhatsApp authentication template; only its hash is stored, with a 5-minute expiry.
4. The employee replies with the code; it is verified and a 15-minute session is opened.
5. `get_employee_details()` reads the employee ID from the session (not from the model) and fetches only allowlisted fields.
6. The reply is sent, sensitive values are redacted in the stored transcript, and every step is written to the audit log.

20.5 Customer Invoice Lookup Example

Illustrative chat (sample data):

```
Customer: Invoice ACC-SINV-2026-00123, my WhatsApp number is +91 98XXXXXX10
AI:       Thanks. I've sent a 6-digit code to that number. Please reply
         with it (valid 5 min).
Customer: 705318
AI:       Verified. Invoice ACC-SINV-2026-00123
         Date: 05 Sep 2026 · Due: 20 Sep 2026 · Status: Partly Paid
```

Total: ₹48,500 · Paid: ₹30,000 · Outstanding: ₹18,500
Delivery: Dispatched

1. The Verification Service confirms the typed number equals the number the chat is coming from.
2. It reads the invoice from ERPNext and checks the invoice contact / customer mobile against that number; a mismatch returns the same generic reply as an invalid invoice.
3. An OTP is sent on WhatsApp and verified; a session is opened bound to this number and this customer.
4. `get_invoice_details()` returns only allowlisted invoice fields for the verified customer.
5. The lookup is logged (fields returned, outcome) without storing the data itself.

22. Data Model Overview

The following core entities make up the relational and vector data model:

Entity	Purpose
users	Platform users (business owners, team members) with authentication credentials.
workspaces	Tenant boundary — isolates each business's data from others.
workspace_members	Links users to workspaces with a role (RBAC).
business_profiles	Business persona configuration: name, description, tone, voice, goals, restrictions.
whatsapp_accounts	Connected WhatsApp Business account credentials and metadata.
contacts	Customer profiles derived from WhatsApp conversations and phone calls.
conversations	A thread of messages between a contact and the business.
messages	Individual inbound/outbound messages within a conversation (text, voice, image, document).
media_assets	Stored voice notes, images, and documents, linked to a message, with transcription/analysis results.
calls	Record of an inbound/outbound phone call: caller, duration, outcome, recording reference.
call_transcripts	Turn-by-turn transcript of a live call between the AI (or human) and the caller.
agents	Definitions of the specialized AI agents available to a workspace.
agent_configs	Configurable parameters for each agent (prompts, thresholds, etc.).
agent_runs	Execution records of an agent acting within a workflow.
knowledge_sources	Registered knowledge inputs — PDF, URL, manual text, FAQ.
documents	Parsed documents derived from a knowledge source.
document_chunks	Chunked, embedded text used for vector similarity search.
leads	Structured lead records captured from conversations or calls.
lead_events	Timeline of updates/events for a given lead.
crm_syncs	Record of each lead's sync status with a CRM provider (provider, external Lead/Contact/Deal ID, last synced, result). UPDATED
crm_connections	Per-workspace CRM provider configuration (Frappe CRM and/or HubSpot), credential references, field mappings, enabled flag.
erp_connections	Self-hosted ERPNext / Frappe site URL, API credential reference, DocType and field allowlist, connectivity status.

otp_challenges	Metadata for each OTP challenge (phone, purpose, subject reference, status, attempts, timestamps). Never the code; its hash lives in Redis with a TTL.
verified_sessions	Short-lived verified identities (employee or customer, subject ID, phone, expiry, revoked flag).
erp_lookup_logs	Audit of each ERP fetch: tool, DocType, field names returned, subject, outcome.
strapi_sources NEW	Connected Strapi instance(s), API token reference, content types synced, last sync status.
marketing_syncs NEW	Record of each lead/contact push to Adobe Marketo Engage or Adobe Campaign, with consent status and result.
outbound_campaigns NEW	Configured proactive campaigns (trigger type, template, channel), with schedule/event binding.
outbound_sends NEW	Log of each outbound WhatsApp message or call sent, its trigger, and delivery/response outcome.
call_sentiment_scores NEW	Per-turn sentiment score for a call, and the resulting trend used for escalation decisions.
erp_write_requests NEW	Drafted and submitted ERP write actions (leave, expense, ticket), verified employee, confirmation status, ERPNext response.
calendar_connections	Connected calendar accounts (Google/Outlook) and their OAuth credentials.
calendar_bookings	Meetings booked by the Scheduling Agent, linked to a lead/conversation/call.
tools	Registry of tools available to agents, with permission metadata.
tool_executions	Audit record of each tool invocation and its result.
human_assignments	Records of conversations or calls assigned to a human during takeover.
webhooks	Configured outbound webhook endpoints (e.g., CRM integration).
webhook_events	Log of outbound webhook deliveries and their status.
prompt_versions	Version history of agent prompts for traceability.
unanswered_questions	Questions the knowledge base could not confidently answer.
audit_logs	System-wide audit trail of security-relevant and business-relevant actions.

23. Backend API Overview

Representative REST endpoints exposed by the FastAPI backend. OTP issue and verification happen inside the chat workflow (internal service calls), not through public endpoints:

Endpoint	Purpose
POST /api/v1/auth/register	Register a new user account.
POST /api/v1/auth/login	Authenticate a user and issue a JWT.
GET /api/v1/conversations	List conversations for the current workspace.
GET /api/v1/conversations/{id}	Retrieve full detail and message history for one conversation.
POST /api/v1/media/analyze	Transcribe a voice note or analyze an image/document.
GET /api/v1/calls	List phone calls for the current workspace.
GET /api/v1/calls/{id}	Retrieve a call's recording, transcript, and outcome.
POST /api/v1/voice/webhook	Receive real-time call events/audio stream from the telephony provider.
POST /api/v1/knowledge/upload	Upload a document (e.g., PDF) into the knowledge base.
POST /api/v1/knowledge/website	Register a website URL as a knowledge source.
GET /api/v1/leads	List captured leads for the current workspace.
POST /api/v1/leads	Manually create a lead record.
POST /api/v1/crm/sync UPDATED	Sync a lead to the workspace's configured CRM provider(s).
GET / PUT /api/v1/integrations/crm	View or change the CRM provider (Frappe CRM, HubSpot, or both) and field mapping.
POST /api/v1/integrations/erpnext/connect	Register the self-hosted ERPNext / Frappe CRM server and API credentials.
POST /api/v1/integrations/erpnext/test	Test connectivity, permissions, and DocType/field allowlist.
GET /api/v1/verification/logs	List OTP challenges and verification outcomes (admin only).
GET /api/v1/erp/lookups	Audit list of ERP look-ups performed by the AI (admin only).
POST /api/v1/knowledge/strapi/connect NEW	Register a Strapi instance, API token, and content types to sync.
POST /api/v1/knowledge/strapi/webhook NEW	Receive a Strapi publish/unpublish event and trigger re-indexing.
POST /api/v1/marketing/adobe/connect NEW	Connect Adobe Marketo Engage or Adobe Campaign via OAuth.
POST /api/v1/marketing/adobe/sync NEW	Push a lead/contact and its behavioral data to the connected Adobe product.
POST /api/v1/campaigns NEW	Create or update a proactive outbound campaign and its trigger.

GET /api/v1/campaigns/{id}/sends NEW	List outbound sends for a campaign and their outcomes.
GET /api/v1/calls/{id}/sentiment NEW	Retrieve the per-turn sentiment trend for a call.
POST /api/v1/erp/requests NEW	Submit a confirmed ERP write action (leave / expense / ticket) for a verified employee.
GET /api/v1/erp/requests NEW	List ERP write actions and their approval status.
POST /api/v1/calendar/connect	Connect a Google/Outlook calendar account via OAuth.
POST /api/v1/calendar/book	Book a calendar event on behalf of a lead/conversation.
GET /api/v1/agents	List configured AI agents for the workspace.
PUT /api/v1/agents/{id}	Update an agent's configuration (persona, prompts, thresholds, voice).
POST /api/v1/playground/chat	Send a test message to the AI in the sandbox Playground.
POST /api/v1/whatsapp/webhook	Receive inbound WhatsApp events from Meta.
GET /api/v1/analytics	Retrieve aggregated analytics and monitoring metrics.

24. Project Folder Structure

The repository follows a modular, service-oriented layout separating backend, frontend, tests, documentation, and deployment concerns:

```
backend/
  agents/           # Supervisor and specialized agent implementations
  workflows/        # LangGraph workflow/state graph definitions
  rag/              # Chunking, embeddings, retrieval pipeline
  media/            # Voice transcription, image/document understanding
  voice/            # Telephony streaming, STT/TTS, live call handling
  whatsapp/         # WhatsApp Cloud API client & webhook handlers
  tools/            # Tool definitions callable by agents
  services/         # Business logic (leads, conversations, etc.)
  database/         # SQLAlchemy models, Alembic migrations
  security/         # Auth, RBAC, permission checks
  integrations/
    crm/            # CRM provider interface + provider registry
    frappe/         # Frappe CRM & ERPNext REST client (read-only ERP)
    hubspot/        # HubSpot CRM provider
    calendar/       # Google/Outlook calendar booking client
    webhooks/       # Generic outbound webhooks
  verification/    # OTP issue/verify, verified sessions, rate limits
  cms/             # Strapi client, publish-webhook handler
  marketing/       # Adobe Experience Cloud (Marketo / Campaign) client
  campaigns/       # Outbound campaign scheduling and trigger evaluation
  sentiment/       # Live call sentiment scoring
  observability/   # Logging, tracing (OpenTelemetry)
  config/          # Environment & settings management

frontend/
  components/      # Reusable UI components
  pages/           # Route-level views
  layouts/         # Page/dashboard layout shells
  hooks/           # Custom React hooks
  services/        # API client layer
  store/           # Zustand client-state stores
  types/           # Shared TypeScript types
  features/        # Feature-scoped modules (leads, inbox, calls, etc.)

tests/            # Backend and frontend test suites
docs/             # Technical documentation
docker/           # Dockerfiles and compose configuration
scripts/          # Developer and operational scripts
```

Separating `agents/`, `workflows/`, `rag/`, `media/`, `voice/`, and `verification/` from generic `services/` keeps AI-specific logic isolated and testable, while `security/` and `observability/` being first-class backend modules reflects the project's production-oriented design goals (Section 29).

25. Development Roadmap

The implementation is planned across thirty-three sequential phases, each with a clear goal, technology focus, and expected output:

Phase	Goal	Main Technologies	Major Tasks	Expected Output
Phase 1	Project Setup	Git, Docker, base configs	Repository scaffolding, environment setup, CI baseline	Runnable empty project skeleton
Phase 2	FastAPI Backend	FastAPI, Pydantic	Core API structure, routing, settings management	Running backend service
Phase 3	PostgreSQL Database	PostgreSQL, SQLAlchemy, Alembic	Schema design, models, migrations	Persistent data layer
Phase 4	React Dashboard	React, TypeScript, Vite, Tailwind	Base UI shell, routing, layout	Navigable dashboard skeleton
Phase 5	Auth & Workspace	JWT, RBAC	User auth, workspace isolation, roles	Multi-tenant login system
Phase 6	WhatsApp Integration	WhatsApp Cloud API, Webhooks	Webhook receiver, message send/receive	Working WhatsApp message loop
Phase 7	OpenAI Integration	OpenAI API	LLM client, prompt handling	Basic AI text responses
Phase 8	LangGraph Supervisor	LangGraph	Supervisor workflow, intent routing	Working agent router
Phase 9	RAG	pgvector, PyMuPDF, embeddings	Ingestion pipeline, chunking, retrieval	Grounded knowledge answers
Phase 10	Specialized Agents	LangGraph, LangChain	Knowledge, Sales, Support, Lead agents	Multi-agent conversation handling
Phase 11	Lead Management	FastAPI, PostgreSQL	Lead extraction, storage, listing UI	Structured lead pipeline
Phase 12	Human Takeover	WebSockets	Live monitoring, pause/resume, assignment	Functional human-in-the-loop system
Phase 13	Tools & Webhooks	Tool framework	Tool registry, permission checks, outbound webhooks	Controlled agent actions
Phase 14	Redis / Background Processing	Redis	Queueing, async workers	Scalable async message handling
Phase 15	Security	JWT, RBAC, validation	Hardening auth, input validation, secrets	Production-grade security baseline

Phase 16	Testing	pytest, Playwright	Unit, integration, and E2E tests	Automated test coverage
Phase 17	Observability	OpenTelemetry, structured logs	Tracing, structured logging	End-to-end request visibility
Phase 18	Docker & Deployment	Docker, Docker Compose	Containerization, deployment scripts	Deployable production build
Phase 19	Media Understanding	Vision model, OCR	Image/document upload handling, analysis pipeline	AI understands images & documents
Phase 20	Voice Messages	Whisper (STT), TTS API	Voice-note transcription, optional TTS replies	Voice notes handled like text
Phase 21	AI Voice Call Agent	Twilio Voice, streaming STT/TTS	Telephony integration, real-time call handling, barge-in	AI answers & holds live phone calls
Phase 22	HubSpot CRM Integration	HubSpot API	OAuth connection, Contact/Deal sync, timeline events	Leads auto-synced to HubSpot
Phase 23	Calendar Booking Automation	Google Calendar / Outlook API	Availability lookup, booking, confirmations	AI books meetings autonomously
Phase 24	CRM Provider Abstraction	Adapter pattern, FastAPI	Provider interface and registry, refactor HubSpot as a provider, per-workspace field mapping	Pluggable CRM layer (Frappe CRM or HubSpot)
Phase 25	Frappe CRM Integration	Frappe REST API	Integration user and API keys, CRM Lead/Deal create and update, de-duplication, notes	Leads sync to self-hosted Frappe CRM
Phase 26	ERPNext Connector	Frappe REST API, httpx	Read-only client, DocType/field allowlist, health checks, retries and circuit breaker	Secure read access to employee, customer, and invoice data
Phase 27	OTP Verification Service	Redis, WhatsApp authentication templates	OTP issue/verify, limits and lockouts, verified sessions, audit	Verified-identity sessions
Phase 28	Employee & Customer Self-Service	LangGraph, ERPNext connector	Enterprise Data Agent, employee flow, invoice flow, masking, tests	OTP-verified ERP look-ups on WhatsApp
Phase 29 NEW	Strapi Knowledge Source	Strapi API, webhooks	Content-type mapping, publish-webhook ingestion, re-indexing	Live, editable knowledge base
Phase 30 NEW	Adobe Experience Cloud Sync	Adobe Marketo / Campaign APIs	OAuth connection, lead/behavioral data mapping, consent gating	Leads flow into marketing automation

Phase 31 NEW	Proactive Outbound Campaigns	WhatsApp templates, Voice Call Agent, background worker	Trigger framework, opt-in/quiet-hours checks, scheduling	AI-initiated reminders and follow-ups
Phase 32 NEW	Live Sentiment Analysis	Streaming transcript scoring	Per-turn scoring, trend detection, escalation trigger	Frustrated callers auto-escalate to a human
Phase 33 NEW	ERP Write Actions	Frappe REST API (write), approval workflow	Scoped write role, confirm-then-submit flow, leave/expense/ticket support	Verified employees can act, not just view

26. MVP vs Advanced Version

MVP — Minimum Viable Product

Only the features required for a working first version:

- Authentication
- Workspace
- WhatsApp integration
- Basic AI response generation
- RAG
- Supervisor Agent
- Knowledge Agent
- Lead capture
- Basic React dashboard
- Conversation history

Advanced Version

- Multiple specialized agents (Sales, Support, Lead Qualification, Human Escalation)
- Human takeover
- Advanced tool calling
- Analytics
- Evaluation framework
- Observability
- Advanced security
- Multi-language support
- Prompt versioning
- Advanced lead scoring
- Automated knowledge updates

Communication Expansion Tier

Building on the Advanced Version, this tier adds the multimodal and voice capabilities covered in this revision:

- Voice message transcription and voice replies
- Image and document understanding
- AI Voice Call Agent — the AI listens to and answers live phone calls
- CRM integration for automatic lead sync (HubSpot in v2; now pluggable with Frappe CRM)
- Calendar booking automation

Enterprise Integration Tier

Building on the Communication Expansion Tier, this tier connects the platform to the company's own systems:

- Pluggable CRM layer — Frappe CRM (self-hosted) or HubSpot, chosen per workspace

-
- Read-only ERPNext connector for employee, customer, and invoice data
 - WhatsApp OTP verification service with verified sessions
 - Employee self-service by Employee ID
 - Customer invoice lookup by Invoice ID and WhatsApp number

Future-Proof Tier NEW

Building on the Enterprise Integration Tier, this tier adds the capabilities in Section 11:

- Strapi-backed dynamic knowledge base
- Adobe Experience Cloud (Marketo Engage / Adobe Campaign) syncing
- Proactive, AI-initiated outbound WhatsApp messages and voice calls
- Live sentiment analysis with automatic human escalation on calls
- Approved ERP write actions (leave, expense claims, support tickets)

27. What Makes This Project Different From a Normal Chatbot?

Traditional Chatbot	This Project
Fixed, scripted responses	LLM-powered, reasoning-based responses
Limited context window / no memory	Stateful conversation memory (LangGraph)
No autonomous workflow	Multi-step agentic workflow with a Supervisor
No specialized agents	Multiple specialized agents (Knowledge, Sales, Lead, Support, Escalation, Voice Call, Scheduling)
Text-only understanding	Multimodal — understands text, voice notes, images, and documents
No phone/voice capability	Answers and holds live phone calls with real-time speech AI
Limited or no CRM integration	Automatic lead sync to self-hosted Frappe CRM or HubSpot for every qualified lead UPDATED
No access to the business's real records	OTP-verified, live look-ups from the company's own ERPNext (employee and invoice data)
Read-only, reactive automation	Verified employees can also write to the ERP (leave, expense, tickets); AI can initiate contact proactively NEW
Static knowledge, uploaded manually	Live, headless-CMS-backed knowledge base (Strapi) NEW
No frustration detection on calls	Live sentiment analysis with automatic escalation NEW
Manual meeting scheduling	Automated calendar availability check & booking
Limited integrations	Tool calling, webhooks, and external system integration
No structured lead extraction	Structured lead qualification and extraction
Limited or no human handoff	Full human-in-the-loop takeover system, including live call transfer
No knowledge grounding	RAG-grounded, source-backed answers
No visibility into decisions	Dashboard visibility, audit logs, observability
No governance over actions	Permission-controlled, validated tool execution

28. Project Capabilities

AI Capabilities

- Natural language understanding
- Intent detection
- Context-aware responses
- Retrieval-Augmented Generation (RAG)

-
- Multi-agent orchestration
 - Multimodal understanding (voice, image, document)
 - Real-time voice conversation (speech-to-text / text-to-speech)
 - Tool calling
 - Conversation memory
 - Multilingual interaction

Business Capabilities

- Customer support
- Lead generation
- Lead qualification
- Sales assistance
- FAQ automation
- Knowledge management
- Human escalation
- Automated meeting scheduling
- Employee self-service, read and write (OTP-verified)
- Customer invoice lookup (OTP-verified)

Automation Capabilities

- WhatsApp automation
- Voice message automation
- AI phone call answering
- Webhooks
- CRM integration — Frappe CRM or HubSpot **UPDATED**
- ERPNext integration (self-hosted)
- WhatsApp OTP identity verification
- Strapi headless CMS knowledge base
- Adobe Experience Cloud marketing sync
- Proactive outbound campaigns
- Live call sentiment analysis
- Calendar booking automation
- Background processing

Management Capabilities

- Dashboard
- Conversation monitoring
- Call monitoring & recordings
- Agent configuration
- Knowledge management
- Analytics
- Audit logs

29. Future Scope

The following capabilities are not part of the current or immediately planned implementation. They are documented here to communicate the platform's long-term direction, and are explicitly out of scope for the phases in Section 24.

- Outbound, AI-initiated proactive voice calls (e.g., appointment reminders, follow-ups) [PLANNED]
- Real-time sentiment analysis during live calls [PLANNED]
- Deeper CRM coverage beyond Leads/Deals (e.g., custom workflows and pipelines in Frappe CRM or HubSpot) [PLANNED]
- Payroll / payslip look-ups with step-up authentication [PLANNED]
- ERP look-ups and write actions over live phone calls with OTP confirmation [PLANNED]
- Expanding ERP write actions beyond leave/expense/tickets (e.g., purchase requisitions) [PLANNED]
- Predictive, AI-scored campaign targeting (who to reach out to and when) [PLANNED]
- Two-way calendar conflict resolution and rescheduling [PLANNED]
- Payment workflows [PLANNED]
- Multi-business SaaS architecture [PLANNED]
- Agent evaluation system [PLANNED]
- Automatic knowledge refresh [PLANNED]
- Advanced analytics [PLANNED]
- AI quality scoring [PLANNED]
- Multi-channel support (Instagram / Telegram / web chat) [PLANNED]

Voice messages, WhatsApp media understanding, image/document understanding, the AI Voice Call Agent, HubSpot CRM integration, and calendar booking automation — previously listed here as future scope — have been promoted into the active architecture as of this revision (see Sections 4, 7, 8, 12, and 18). Frappe CRM and ERPNext integration with OTP-verified self-service is likewise part of the active scope as of v3 (see Sections 9 and 10). Proactive outbound campaigns and live call sentiment analysis, previously listed here, are part of the active scope as of v4 (see Section 11).

30. Technical Quality Requirements

This project is being designed as a production-oriented system, not a simple college or hobby chatbot project. The following engineering practices are treated as requirements, not optional polish:

- Modular architecture
- Separation of concerns
- API-first design
- Database migrations
- Type safety (TypeScript, Pydantic, type hints)
- Input validation at every boundary
- Automated testing
- Security by design
- Observability

-
- Robust error handling
 - Structured logging
 - Auditability
 - Background processing for long-running work
 - Scalable architecture
 - Human-in-the-loop safeguards
 - Controlled, permissioned tool execution
 - Low-latency, streaming-first design for real-time voice calls
 - Privacy by design: data minimisation, verified access, and masked logging for ERP and CRM data
 - Resilient integration with self-hosted systems: timeouts, retries, circuit breakers, graceful degradation

31. Important Terminology

The following terms are used throughout this document, defined briefly on first reference:

Agentic AI

An AI system that plans, decides, and acts autonomously across multiple steps, rather than producing a single one-shot response.

Multi-Agent System

An architecture composed of several specialized AI agents, each responsible for a distinct capability, coordinated by a supervisor.

Supervisor Agent

The coordinating agent responsible for intent detection, routing, and overall workflow control.

RAG (Retrieval-Augmented Generation)

A technique that retrieves relevant information from a knowledge base and supplies it to an LLM before generating a response, to ground answers in real data.

Vector Search

Searching for semantically similar content by comparing numerical vector representations (embeddings) rather than exact keyword matches.

Embeddings

Numerical vector representations of text that capture semantic meaning, used for similarity search.

Tool Calling

An LLM's ability to invoke predefined functions/tools to take real actions, instead of only producing text.

Stateful Workflow

A process that persists and tracks its own state (e.g., conversation progress) across multiple steps or interactions.

Human-in-the-Loop

A design pattern where a human can intervene in or take over an automated process at defined points.

LLM (Large Language Model)

A machine learning model trained on large volumes of text, capable of understanding and generating natural language.

Multimodal

The ability of a model or system to understand more than one type of input — text, images, documents, and audio — rather than text alone.

STT (Speech-to-Text)

The process of converting spoken audio into a written transcript.

TTS (Text-to-Speech)

The process of converting written text into natural-sounding spoken audio.

Streaming Voice / Real-Time Voice

Processing audio continuously as it arrives, so a system can listen and respond during a live conversation rather than waiting for the whole recording to finish.

Barge-In

The ability for a caller to interrupt the AI's spoken response mid-sentence, as in a natural human conversation.

Telephony Provider

A third-party service (e.g., Twilio) that connects an application to the public phone network for making and receiving calls.

Webhook

An HTTP callback that delivers real-time event data from one system to another as events occur.

REST API

An architectural style for building web APIs using standard HTTP methods and resource-oriented URLs.

WebSocket

A protocol enabling persistent, two-way, real-time communication between a client and a server.

Semantic Search

Search based on meaning/context rather than exact keyword matching.

Lead Qualification

The process of determining whether a potential customer has genuine intent, budget, and fit to proceed.

Knowledge Base

The structured collection of business information (documents, FAQs, product data) available to the AI.

Prompt Engineering

The practice of designing and refining the instructions given to an LLM to produce reliable, desired behavior.

Observability

The ability to understand a system's internal state from its external outputs — logs, metrics, and traces.

RBAC (Role-Based Access Control)

An access-control model where permissions are assigned based on user roles.

Audit Logging

Recording a chronological trail of system actions and decisions for accountability and review.

Workspace Isolation

Ensuring each business's data and configuration are kept separate from other businesses (multi-tenancy).

CRM (Customer Relationship Management)

A system, such as Frappe CRM or HubSpot, used to store and manage customer and lead records for a sales team.

Frappe Framework

The open-source Python/JavaScript web framework on which ERPNext, Frappe HR, and Frappe CRM are built; it provides DocTypes, REST APIs, permissions, and webhooks.

ERPNext

An open-source ERP system built on Frappe, covering accounting, invoicing, customers, sales, purchasing, inventory, and more.

Frappe CRM

An open-source CRM built on Frappe for managing leads, deals, contacts, and organizations.

Frappe HR (HRMS)

The HR and payroll application for the Frappe ecosystem (leave, attendance, payroll, expense claims).

DocType

Frappe's record type (for example Employee or Sales Invoice); each DocType has fields, permissions, and an automatic REST API.

OTP (One-Time Password)

A short-lived, single-use code sent to a person to confirm they control a given phone number at that moment.

Authentication Template

A WhatsApp message template category designed for delivering OTPs and verification codes.

Verified Session

A short-lived, server-side record that a specific person has passed verification, used to authorise identity-bound tools.

Step-Up Authentication

An additional verification required before more sensitive data is shown.

Provider Interface (Adapter Pattern)

A common interface that lets different back-end systems, such as Frappe CRM and HubSpot, be swapped without changing the agents that use them.

Anti-Enumeration

Returning identical responses for different failure causes so attackers cannot discover which IDs or numbers exist.

Headless CMS

NEW

A content management system, such as Strapi, that stores and serves content through an API rather than rendering its own website — letting other systems, like this AI, consume the content directly.

Adobe Experience Cloud

NEW

Adobe's suite of marketing products, including Marketo Engage (B2B lead nurturing) and Adobe Campaign (cross-channel campaign management).

Proactive / Outbound Messaging

NEW

A message or call the AI initiates on its own trigger (a schedule or an event), rather than one sent in reply to the customer.

Sentiment Analysis

NEW

Automatically judging the emotional tone (e.g., positive, neutral, frustrated) of text or speech.

Confirm-Then-Submit

NEW

A safety pattern where the AI restates the action it is about to take and waits for explicit confirmation before submitting it.

Write Action

NEW

An operation that creates or changes a record in an external system (as opposed to a read-only look-up), such as filing a leave application.

32. Final Executive Summary

What are we building?

An AI-powered WhatsApp and voice-call automation platform that connects a business's WhatsApp communication and phone line to an agentic AI system capable of understanding customers — through text, voice notes, images, documents, and live calls — answering questions from the business's own knowledge base, qualifying leads, booking meetings, securely serving verified employee and customer information — and now accepting approved write requests such as leave applications — from the company's own ERPNext and Frappe CRM, keeping its own knowledge current through Strapi, feeding Adobe Experience Cloud for marketing automation, reaching out proactively, and escalating to a human automatically when a caller becomes frustrated or needs a person.

Why are we building it?

Businesses using WhatsApp and phone lines face repetitive questions, slow response times, missed calls, lost leads, inconsistent answers, scattered knowledge, manual scheduling, and no centralized visibility into their customer conversations. This platform solves those problems with grounded, monitored, multimodal automation.

What can it do?

It understands intent from text, voice notes, images, and documents; answers and holds live phone calls; answers questions using RAG-grounded knowledge; qualifies and captures structured leads and syncs them to Frappe CRM or HubSpot; books meetings automatically on a connected calendar; lets employees and customers retrieve their own ERPNext details (employee profile, invoice status) in chat after WhatsApp OTP verification, and lets verified employees submit leave, expense, and ticket requests back into ERPNext; executes controlled tools/actions; maintains conversation context; escalates to humans on chat or live calls; and exposes everything through a React dashboard.

Why is it Agentic AI?

A Supervisor Agent coordinates specialized agents (Knowledge, Sales, Lead Qualification, Support, Human Escalation, Media Understanding, Voice Call, Scheduling), uses stateful LangGraph workflows, calls permissioned tools, understands multiple media types, and supports human-in-the-loop intervention — behavior far beyond a scripted chatbot.

What technologies are being used?

React/TypeScript on the frontend; FastAPI/Python on the backend; OpenAI and LangGraph for AI orchestration; OpenAI Whisper and TTS for voice; Twilio Voice for telephony; PostgreSQL with pgvector for data and vector search; Redis for async processing; WebSockets for real-time updates; Frappe CRM (self-hosted) or HubSpot for CRM; ERPNext for employee and invoice data (read and approved writes); Strapi for a live, editable knowledge base; Adobe Marketo Engage / Adobe Campaign for marketing sync; WhatsApp OTP for identity verification; Google Calendar/Outlook for scheduling; and Docker for deployment — full detail in Section 18.

What is the overall architecture?

Messages flow from WhatsApp through FastAPI and a Redis queue into a LangGraph-based agent workflow, coordinated by a Supervisor Agent that routes to specialized agents backed by RAG (now including a live Strapi knowledge source), tools, a pluggable CRM layer (Frappe CRM or HubSpot),

self-hosted ERPNext (read and approved writes), Adobe Experience Cloud, and calendar integrations, on a PostgreSQL/pgvector database. Any request for personal or financial data, or any ERP write action, passes through a deterministic OTP verification service before the connector layer is reached, and a confirm-then-submit step before anything is written. An Outbound Campaign Agent can also initiate contact, and a Sentiment Monitoring Agent watches live calls for frustration and triggers human escalation. In parallel, live phone calls stream through a telephony provider into the same Supervisor Agent via real-time speech-to-text and text-to-speech — with the React dashboard connected via WebSockets for real-time monitoring and human takeover of either channel.

What are the major future capabilities?

Payroll/payslip look-ups with step-up authentication, ERP look-ups and write actions over live phone calls, expanding write actions beyond leave/expense/tickets, predictive campaign targeting, deeper CRM workflow coverage, payment workflows, a multi-business SaaS model, automated knowledge refresh, an agent evaluation framework, and multi-channel support beyond WhatsApp (see Section 28, Future Scope).